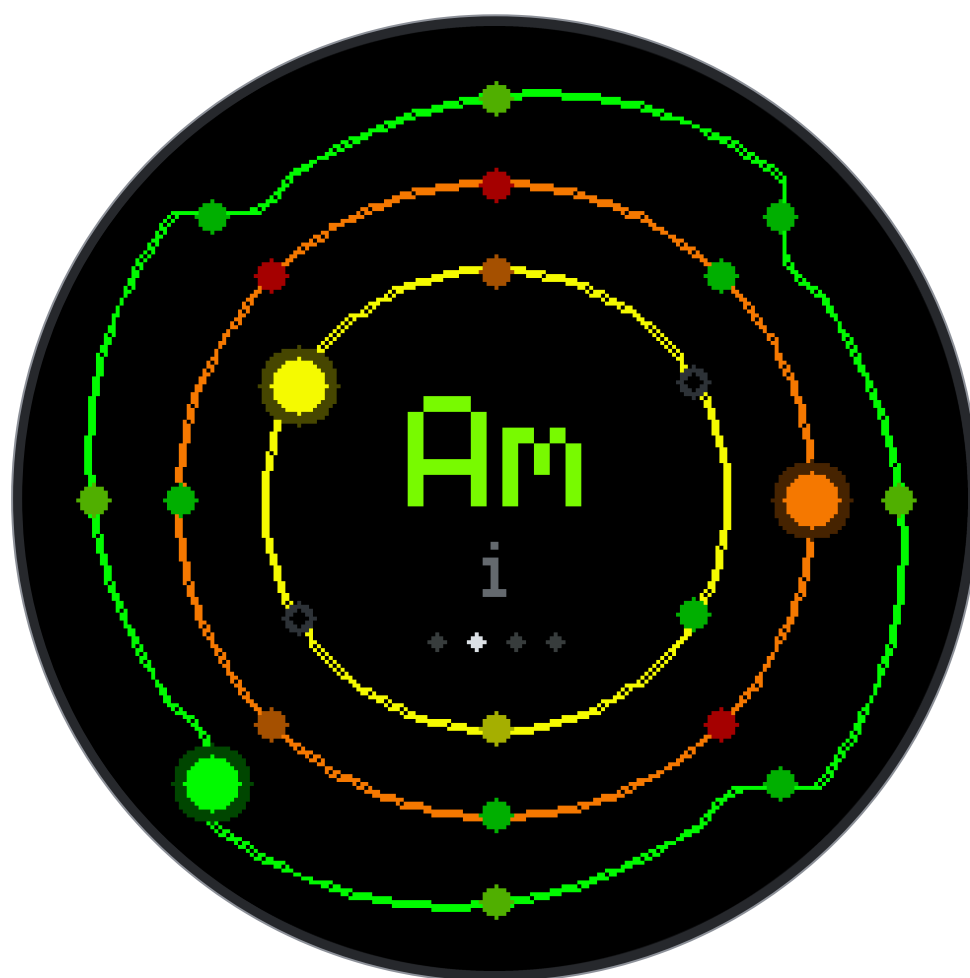


OWNER'S MANUAL

AMMONITE

Three-voice arpeggiator synthesizer



CONTENTS

Every screen picture in this manual comes straight from Ammonite's own code: it is exactly what the round screen shows. A word on words: **page** always means one of Ammonite's nine knob pages (*page 3 ARP*); **p. 12** means a page of this booklet.

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START HERE

Quick start

Ammonite plays the moment it has power. Five minutes from here you will have changed its key, its tempo and its chords.

- 1 Turn it down, then plug in.** Turn knob 12 (VOLUME) to the left, connect the stereo output to your amp, mixer or speakers, and power Ammonite from USB. Then bring VOLUME up.
- 2 Listen.** Three arpeggios in A minor at 110 BPM. OSC 1 plays the low root and fifth in quarter notes, OSC 2 runs up and down the chord in eighths, OSC 3 picks random notes of the scale in sixteenths, leaving a few steps out. No effects are on yet.

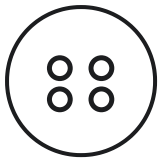


Power-on. Knob 11 points at page 1, MAIN. The screen shows the three arps as rings (see *Reading the screen*, p. 23).

- 3 Change the tempo.** Turn knob 6, TEMPO. The screen shows what you are turning and its value, then fades back to the rings.
- 4 Change the key.** Knob 4 is ROOT, knob 5 is SCALE. Try DORIAN or PENTA. All three arps follow at once; nothing can play a wrong note.
- 5 Let the chords move.** Turn knob 11 all the way right, to page 9, KEY. Turn knob 1 (PROG) to POP. Now the chord changes every bar, and every arp follows it.
- 6 Add echoes and space.** Knob 11 to page 7, DELAY: turn up knobs 7–9, the delay SEND of each oscillator. Then page 8, MIX: knobs 7–9 are the reverb sends.

TWO THINGS WORTH KNOWING NOW

- **Changing page never changes the sound.** But when you then touch a knob, its setting jumps to where the knob points. The whole story is on p. 4.
- **Nothing is saved.** Switch off and on, and Ammonite starts again from its power-on sound, with one exception: the page knob 11 points at when you switch on takes its settings from where its knobs are. With knob 11 on page 1, that means the levels, key, tempo and masters start where you left the knobs.



TWELVE KNOBS, NINE PAGES

The panel

Twelve knobs control about a hundred settings. They are organised so that once you know four rules, you can find anything.



on pages 2 to 8, each column is one oscillator

shared setting
or SECTION

PAGE
1 to 9

VOLUME
on every page

1. Knob 11 picks the page

There are nine pages. Turn knob 11 and the screen lists the new page's knobs. Knob 12 is always VOLUME, whatever the page.

PAGE	WHAT ITS KNOBS DO	KNOB 10	MANUAL
1 MAIN	levels, key, tempo, and master cutoff, delay and reverb	SWING	p. 8
2 OSC	each oscillator's shape, octave and starting note	DETUNE	p. 10
3 ARP	what each arp plays and when	SECTION	p. 11
4 ENVELOPE	how each note starts and ends, in volume, filter and pitch	SECTION	p. 13
5 FILTER	brightness and tone of each oscillator	DRIVE	p. 15
6 LFO	slow (or fast) movement of pitch, cutoff, volume, pan, shape	TARGET	p. 16
7 DELAY	each oscillator's own echo	SECTION	p. 18
8 MIX	pan, stereo width and reverb send	SIZE	p. 20
9 KEY	chord progressions, tuning, reverb tone, the screen view	VIEW	p. 21

2. Columns are oscillators

On pages 2 to 8, each column of knobs belongs to one oscillator, in both groups: knobs 1 4 7 are OSC 1, 2 5 8 are OSC 2, 3 6 9 are OSC 3. Each row does one job for all three: on page 5, for example, knobs 1–3 are the three CUTOFFS, 4–6 the three RESOs and 7–9 the three TYPES. The screen lists a page the same way: 1-3 CUTOFF, 4-6 RESO...

Pages 1 MAIN and 9 KEY are different: they set things for the whole instrument, so every knob there has its own job.

3. Knob 10 is shared, or picks a section

On most pages knob 10 is one setting shared by all three oscillators (DETUNE, DRIVE, SIZE...). On four pages there is more to control than nine knobs can hold, so the page is split into **sections** and knob 10 chooses one: ARP, ENVELOPE and DELAY call it SECTION, LFO calls it TARGET. Choosing a section only changes what knobs 1–9 control. It never changes the sound.

4. A knob takes over when you turn it

Every knob on every page and section remembers its own setting. When you change page (or section) the knobs themselves stay where they are, so most of them now point somewhere other than the setting they control. Ammonite leaves every setting alone until you turn its knob. The moment you do, the setting **jumps** to where the knob points, and from then on follows it.

So glance at a knob's pointer before you turn it. If it points far from where the sound is, expect a jump. A tiny wobble does not count as a turn, so brushing a knob does nothing.

The screen tells you what you touched



Readout. Turn any knob: its name, value and a gauge. Column pages add the oscillator (and section) above the name.

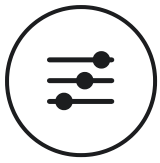


Page map. Turn knob 11: the page's name and what each row of knobs does.



Section map. Turn knob 10 on a page with sections: the section, its rows, and one dot per section.

About 2.5 seconds after you let go, the text fades and the rings return. Under a section map, the bright dot is the section you are in. On the LFO page, a colored dot marks a target that is doing something (its DEPTH is up on some oscillator) and a grey dot one that is off.



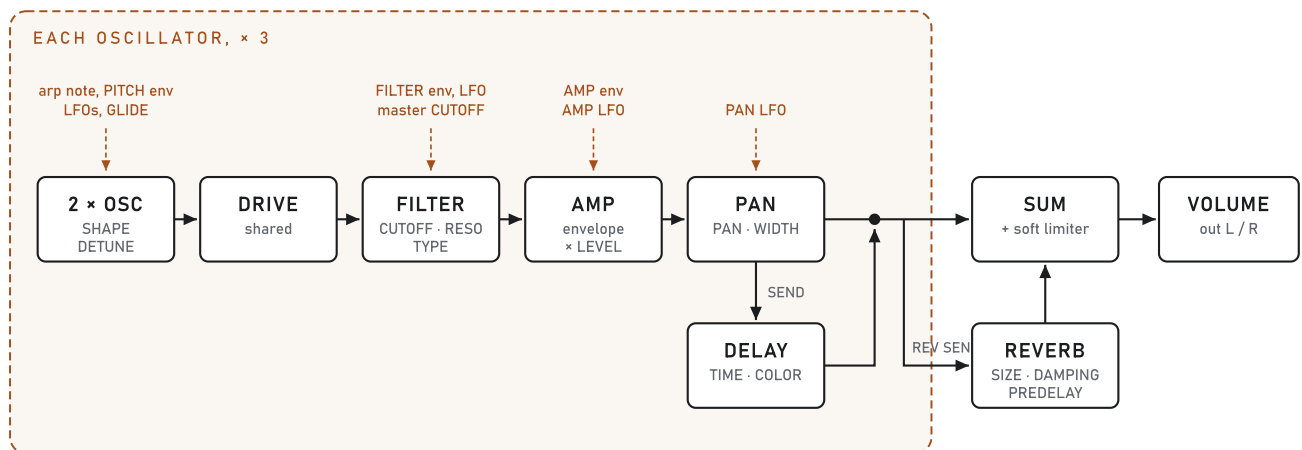
THE IDEAS BEHIND THE KNOBS

How Ammonite plays

Three voices, one key and one clock. Everything on the nine pages is a detail of one of these three.

Three voices

Ammonite has three oscillators, OSC 1 to OSC 3. Each is a complete little synth with its own arpeggiator, envelopes, filter, LFOs, echo and place in the stereo mix. They play one note at a time each, so together they make three-part music. Inside, every oscillator is really *two* oscillators, pulled slightly apart by DETUNE for a fuller sound.



One key

You never choose notes on Ammonite. You choose a **key** (ROOT and SCALE on page 1), a **chord** within it, and then, for each oscillator, which of the chord's notes it may use. That set of notes is the oscillator's **pool**, and its arp plays from nothing else. Here is how the power-on sound gets its notes:

Key

ROOT A, SCALE MINOR



Chord

i, the first chord: Am



OSC 1 pool

DEGREE +0, FIFTHS, 1 octave



OSC 2 pool

DEGREE +2, TRIAD, 2 octaves



OSC 3 pool

DEGREE +4, KEY, 2 octaves

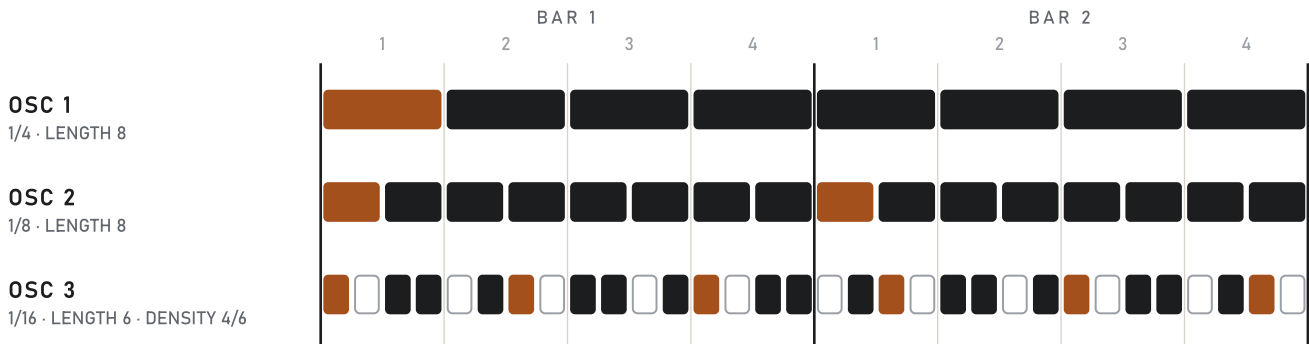


DEGREE (page 2) moves a pool's starting note up the scale from the chord's root: OSC 2 starts two scale steps up, on C, so its TRIAD is C E G. POOL and RANGE (page 3) say which notes to take and over how many octaves; OCTAVE (page 2) places the pool high or low. The colored dots are each note's screen color (p. 23).

When the chord changes, from a progression or from CHORD on page 9, every pool is rebuilt on the new chord, and all three oscillators move with it. SCALE can also be a **chord type** (MAJ, MIN7, ADD9...) instead of a scale: then the key is that chord, and a progression moves it around, keeping its color.

One clock

A single master clock (TEMPO on page 1) drives all three arps, the progression, the delay times and, if you want, the LFOs. Each arp divides the beat its own way (DIVISION), but every grid starts from the same beat, so however strange the combination, the three never drift apart.



Two bars of the power-on patterns. Each block is a step; rust marks the first step of a pattern. OSC 3's pattern is 6 steps long, so it starts over at different places in each bar; its DENSITY of 4/6 leaves out the same two steps every time round (hollow).

Three things shape the rhythm beyond DIVISION. **LENGTH** sets how many steps a pattern has before it starts over; different lengths make patterns that slide against each other. **DENSITY** leaves steps out, spread as evenly as possible. **SWING** (page 1, knob 10) delays every second step. And RESTART on page 9 can pull every pattern back to its first step on each bar or each chord.

Arps and drones

Set an oscillator's MODE (page 3) to OFF and it stops arpeggiating: it holds a single note (the first of its pool) and slides to the new note when the chord changes. A drone ignores the amp envelope's decay: it fades in over ATTACK and then simply holds. One drone under two arps, or three drones and no arps at all, are both good ways to play Ammonite.

1

MAIN

The performance page: how loud each oscillator is, what key and tempo you are in, and three master knobs that move a whole part of the sound at once.



Page 1 just chosen. MAIN is a whole-instrument page: every knob has its own job. The screen's two columns mirror the panel's two groups.

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	LEVEL	0 to 100	80, 70, 60	The volume of OSC 1, 2 and 3.
4	ROOT	C to B	A	The key's home note. Everything transposes with it, progressions included.
5	SCALE	10 scales, then 10 chord types	MINOR	Which notes belong to the key. Turn past BLUES for chord keys, MAJ to ADD9: the key becomes that chord. Every setting is spelled out on p. 30.
6	TEMPO	40 to 240 BPM	110 BPM	The master clock: arps, chord changes, delay times and synced LFOs.
7	CUTOFF	-3.0 to +3.0 OCT	+0.0 OCT	Moves all three filters together, up to three octaves either way from where page 5 has them. A performance filter sweep.
8	DELAY	0.0 to 2.0 X	1.0 X	Scales all three delay SENDs (page 7) at once: 0 silences every echo, 2 doubles every send.
9	REVERB	0.0 to 2.0 X	1.0 X	The same for the three reverb sends (page 8).
10	SWING	0 to 100	0	Delays every second step of every arp. At 100 the steps go long-short, 2 to 1: a full triplet shuffle.

Why the masters sit in the middle. The three master knobs multiply or shift what the other pages set. At their centre positions they change nothing, so you can dial in each oscillator's cutoff, echo and reverb carefully on pages 5, 7 and 8, then come back here and move all three together while you play.

TRY THIS

- Turn SCALE slowly from one end to the other while the arps run. Each step is a new key; the chord keys at the far end are the most colorful.
- Sweep CUTOFF down to -3 and back for a whole-mix filter build.
- SWING at about 30 loosens sixteenths without sounding like a shuffle.



TEMPO turned to 120 BPM. Page 1 has no columns, so the readout shows no oscillator.

2

OSC

What each oscillator sounds like, how high it plays, and which note of the chord it starts from.



Page 2. Row A (knobs 1–3) is SHAPE, row B (4–6) OCTAVE, row C (7–9) DEGREE; knob 10 is DETUNE, shared by all three.

KNOBS	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	SHAPE	0 to 100	100, 50, 30	The waveform, morphing smoothly: 0 a pure sine (soft, round), 50 a square (hollow, woody), 100 a saw (bright, buzzy).
4 5 6	OCTAVE	–1 to +3	+0, +1, +2	Moves the oscillator's whole pool by octaves. At +0 the pool starts around the second octave (A2 at power-on); each step doubles the pitch.
7 8 9	DEGREE	+0 to +7	+0, +2, +4	Where the pool starts, in steps up the scale from the chord's root. The power-on values are the chord's root, third and fifth: with every arp off, the three drone the chord.
10	DETUNE	0.0 to 20.0 CT	8.0 CT	Each oscillator is two oscillators; DETUNE pulls them apart (in cents, hundredths of a semitone). A little gives a gentle chorus; 20 is wide and wobbly. MIX WIDTH (page 8) can spread the two apart in stereo.

TRY THIS

- **A bass line.** OSC 1 at SHAPE 0 and OCTAVE –1 gives a round sub bass under the other two.
- **Unison.** Give OSC 2 and OSC 3 the same DEGREE and OCTAVE as OSC 1, but different MODEs on page 3: three rhythms on the same notes.
- **Chord colors.** With every arp off, try DEGREES 0, 2, 4, 6: a seventh chord that follows the progression.

3

ARP

Which notes each oscillator plays, when it plays them, and how often it surprises you.

Every oscillator has its own arpeggiator. On each step it takes the next note from its *pool* (p. 6), a handful of notes built from the key and the chord playing now. All three arps run from one master clock, so however different their rhythms, they never drift apart.

There are too many settings for one page of knobs, so ARP has three sections. Knob 10 picks the section. Choosing a section only changes what knobs 1–9 control, never the sound.



The ARP page with the **CHANCE** section chosen. Knob 10 has just been turned, so the screen lists the section's knobs. Each column is one oscillator: knobs 1, 4 and 7 set OSC 1; knobs 2, 5 and 8 set OSC 2; knobs 3, 6 and 9 set OSC 3.



NOTES

which notes



RHYTHM

when they play



CHANCE

how often, how surprising

NOTES

KNOBS	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	MODE	OFF LOOP UP DOWN UP-DN RANDOM	UP, UP-DN, RANDOM	The order the arp walks through its pool. LOOP repeats one note in rhythm. OFF stops the arp: the oscillator holds a single note, a drone that slides to follow the chord.
4 5 6	POOL	KEY TRIAD 7TH FIFTHS ROOT	FIFTHS, TRIAD, KEY	Which notes the arp may use. KEY: every note of the scale. TRIAD and 7TH: the chord's three or four notes. FIFTHS: root and fifth. ROOT: the root alone, in octaves.
7 8 9	RANGE	1 OCT 2 OCT 3 OCT	1, 2, 2 OCT	How many octaves the pool spans.

Where a pool starts is set on page 2, OSC: **DEGREE** moves it up the scale from the chord's root, **OCTAVE** moves it by octaves.

RHYTHM

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	DIVISION	1/1 1/2 1/4 1/4T 1/8 1/8T 1/16 1/16T 1/32	1/4, 1/8, 1/16	The length of one step, as a note value of the tempo. T is a triplet: three steps in the time of two.
4 5 6	LENGTH	1 to 16 steps	8, 8, 6	How many steps before the pattern starts over. Give the oscillators different lengths and their patterns slide against each other, lining up again only now and then.
7 8 9	GATE	0 to 100	50	How long each note is held, as a share of its step: short for staccato, full for legato. It matters most when the amp envelope has some SUSTAIN (page 4).

CHANCE

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	DENSITY	none to every step, shown as 3/8	8/8, 8/8, 4/6	How many of the pattern's steps play. The hits are spread as evenly as possible (a <i>Euclidean</i> rhythm), and the same steps play every time round: a pattern, not a dice roll.
4 5 6	VARY	0 to 100	0	The chance that a step plays a random note from its pool instead of the next one.
7 8 9	RATCHET	0 to 100	0	The chance that a step splits into 2 to 4 quick repeats of its note.



Turning a knob shows its readout: section, oscillator, name and value. Here OSC 3's DENSITY is at 3/6: three of its six steps play.



A few seconds later the readout fades. The inner ring (OSC 3) now shows its three rests as hollow dots, spaced evenly around the ring.

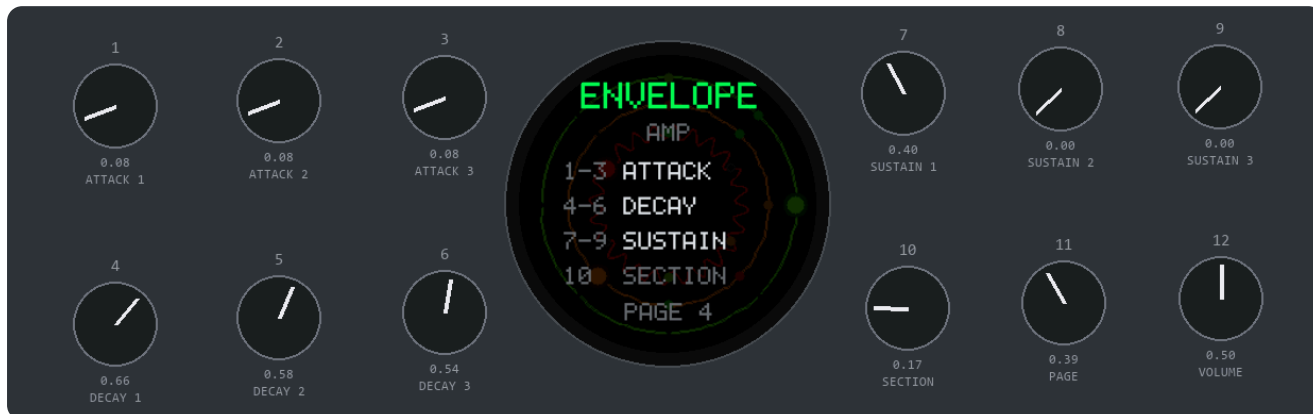
TRY THIS

- **A classic off-beat.** On OSC 1 set **LENGTH** 8 and **DENSITY** 3/8: the *tresillo*, the 3-3-2 rhythm under half of all pop music.
- **Patterns that drift.** Set **DIVISION** to 1/8 on all three oscillators and **LENGTH** to 8, 7 and 5. The three patterns only line up again after 280 steps.
- **A little life.** On the fastest oscillator, raise **VARY** to about 20 and **RATCHET** to about 10.

4

ENVELOPE

What happens inside each note: how its volume rises and falls, how the filter opens with it, and how its pitch arrives.

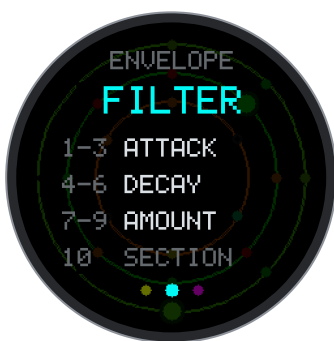


Page 4 just chosen, AMP section. Knob 10 switches between AMP, FILTER and PITCH.



AMP

volume of each note



FILTER

a filter sweep per note



PITCH

how each note arrives

AMP

KNOBS	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	ATTACK	1 MS to 4.1 S	2 MS	How long a note takes to reach full volume. Short is a click-free hit; long is a swell.
4 5 6	DECAY	5 MS to 4.0 S	401, 249, 180 MS	How long it takes to fall to the SUSTAIN level, and also how long the note takes to fade out once its GATE ends.
7 8 9	SUSTAIN	0 to 100	40, 0, 0	The level a note holds while its GATE (page 3) lasts. 0 makes every note a pluck that dies away over DECAY.

A new note starts from wherever the last one had got to, so fast patterns never click. An oscillator whose MODE is OFF (a drone) fades in over ATTACK and then holds.

FILTER

KNOBS	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	ATTACK	1 MS to 4.1 S	2 MS	How long the filter takes to open to its peak after each note starts.
4 5 6	DECAY	5 MS to 4.0 S	249 MS	How long it takes to close back to the CUTOFF set on page 5.
7 8 9	AMOUNT	-4.0 to +4.0 OCT	+0.0 OCT	How far each note moves the cutoff. Positive opens it (brighter at the start of the note), negative closes it (the note starts dark and opens up). 0 switches it off.

PITCH

KNOBS	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	AMOUNT	-12 to +12 ST	+0 ST	Each note starts this many semitones away from its pitch and slides into place.
4 5 6	DECAY	5 MS to 997 MS	24 MS	How long that slide takes. A few milliseconds makes a click or a bell-like ping; longer makes a swoop.
7 8 9	GLIDE	0 MS to 1.0 S	0 MS	Portamento: each arp note slides from the last one instead of jumping. Drones always slide a little (about 0.4 s) when the chord changes; a longer GLIDE slows that too.

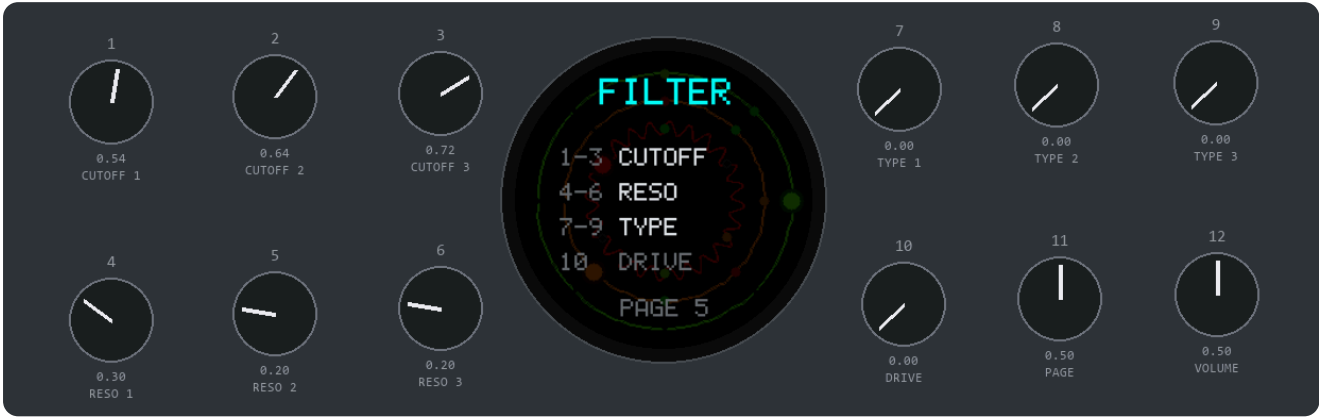
TRY THIS

- **Pads.** ATTACK around 1 S, SUSTAIN 80, GATE 100 on page 3, and a slow DIVISION such as 1/2.
- **Squelch.** FILTER AMOUNT +3.0 OCT, FILTER DECAY about 150 MS, with CUTOFF low and RESO high on page 5.
- **Bells.** SHAPE 0 on page 2, PITCH AMOUNT +12 ST, PITCH DECAY 8 MS, AMP DECAY about 900 MS. See the recipe on p. 25.

5

FILTER

Each oscillator has its own filter: the main way to make it brighter or darker, thinner or rounder.



Page 5. CUTOFF, RESO and TYPE per oscillator; knob 10 is DRIVE, shared.

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	CUTOFF	80 HZ to 12.0 KHZ	1.2, 2.0, 3.0 KHZ	The filter's frequency. For the low-pass type: lower is darker and softer.
4 5 6	RESO	0 to 100	30, 20, 20	Resonance: a peak at the cutoff. Higher values whistle and sing, especially when the cutoff moves.
7 8 9	TYPE	LP LP-BP BP BP-HP HP	LP	A smooth morph from low-pass (keeps the lows) through band-pass (a narrow, nasal band) to high-pass (keeps the highs, thin and airy).
10	DRIVE	0 to 100	0	Pushes all three oscillators into soft saturation before their filters: warmer, then grittier. The level stays roughly the same.

Four things move the cutoff. The CUTOFF knob here sets where each filter rests. On top of it come the master CUTOFF on page 1 (all three at once), the FILTER envelope on page 4 (with every note) and the CUTOFF LFO on page 6 (slowly, or not so slowly). They add up, so a filter can move a long way.

TRY THIS

- TYPE BP with RESO around 60 turns a saw into a vowel-like, talking sound.
- Put OSC 3 on HP: its high arp becomes a thin shimmer over the others.



Knob 2 turned: OSC 2's CUTOFF at 1.5 KHZ.

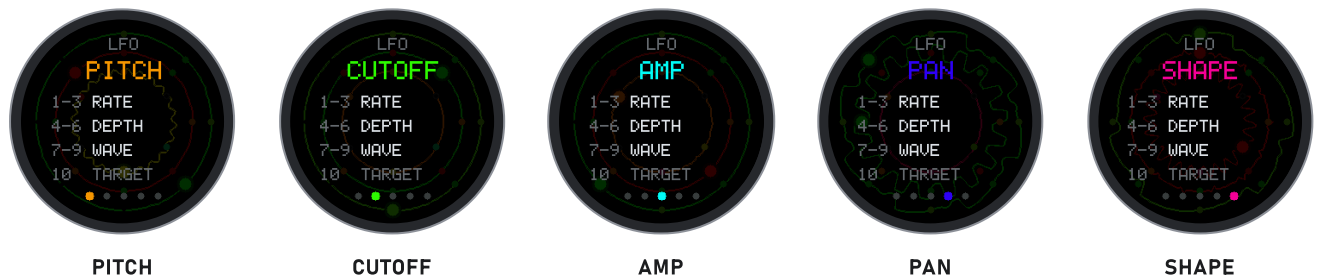
6

LFO

Movement. Each oscillator has five LFOs, one for each thing they can move: pitch, cutoff, volume, pan and shape. That is fifteen in all, each with its own speed.



Page 6, PITCH target. Knob 10 is called TARGET here: it picks what the LFOs on knobs 1–9 move.



Every target has the same three rows. At power-on every DEPTH is 0, so the dots of the other targets are grey: an LFO does nothing until you raise its DEPTH.

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	RATE	0.02 to 20.0 HZ; with LFO SYNC on TEMPO: 4 BARS 2 BARS 1 BAR 1/2 1/4 1/4T 1/8 1/8T 1/16	0.32 HZ (CUTOFF: 0.09, 0.13, 0.17 HZ)	How fast the LFO cycles: from about once a minute to a buzz. With LFO SYNC (page 9) set to TEMPO, RATE picks a note length instead, and the LFO locks to the bar.
4 5 6	DEPTH	0 to 100	0	How much it moves its target (table below).
7 8 9	WAVE	SINE TRI SAW SQUARE S+H DRIFT	SINE	The movement's shape. S+H jumps to a new random value each cycle; DRIFT wanders smoothly from one random value to the next.

TARGET	AT DEPTH 100	SOUNDS LIKE
PITCH	up and down one semitone	vibrato, or a seasick wobble when slow
CUTOFF	three octaves either way	a filter that breathes or wah-wahs
AMP	dips all the way to silence (never louder)	tremolo, or a chopped gate with SQUARE
PAN	swings from full left to full right	notes moving around the room
SHAPE	50 either side of the SHAPE setting	a tone that keeps changing color

TRY THIS

- **Slow life.** CUTOFF target, DEPTH about 30 on all three: the power-on rates are slow and all different, so the three filters drift in and out of step.
- **Rhythmic chop.** Set LFO SYNC to TEMPO on page 9, then AMP target, RATE 1/8, WAVE SQUARE, DEPTH 100 on a drone.
- **Wide arps.** PAN target, WAVE S+H, RATE about 4 HZ: every few notes land somewhere new in the stereo field.

7

DELAY

Each oscillator has its own stereo echo, timed to the tempo. Three delays, three different rhythms of repeats.



Page 7, TIME section. Knob 10 switches between TIME and COLOR. All the SENDs start at 0: turn one up to hear its echo.



TIME

when the echoes come



COLOR

what they sound like

Each echo is fed from its oscillator after the pan, so it starts in the same place in the stereo field. The echoes also go on to the reverb with the dry sound. The master DELAY knob on page 1 scales all three SENDs at once.

TIME

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	DIVISION	1/32 1/16 1/16T 1/8 1/8T 1/8. 1/4 1/4T 1/4. 1/2 1/1	1/8., 1/4, 1/8T	The time between echoes as a note value of the tempo. A dot (1/8.) is one and a half times as long: the classic dotted echo that bounces off the beat.
4 5 6	FEEDBACK	0 to 100	40	How much of each echo comes back again: a single repeat at 0, a long tail near 100.
7 8 9	SEND	0 to 100	0	How much of the oscillator goes into its delay. At 0 the delay is off.

COLOR

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	TONE	500 HZ to 12.0 KHZ	4.0 KHZ	Each repeat passes through a gentle low-pass: lower settings make the echoes darker and darker as they fade, like tape.
4 5 6	WOBBLE	0.0 to 3.0 MS	0.9 MS	A slow, random wander in the delay time, the way a tape machine drifts. It bends the pitch of the echoes slightly.
7 8 9	WIDTH	0 to 100	100	At 0 the echoes stay where the oscillator is panned. At 100 they ping-pong: left, right, left.

TRY THIS

- Delay only the slowest oscillator, with a dotted DIVISION: the echoes fill the gaps between its notes.
- FEEDBACK 80, TONE 1.5 KHZ, WOBBLE 2 MS: a dark dub echo that sinks away (see the recipe *Dub echoes*).

MIX

Where each oscillator sits in the stereo picture, and how much of it goes to the reverb.



Page 8. PAN, WIDTH and REV SEND per oscillator; knob 10 is the reverb's SIZE.

KNOBS	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1 2 3	PAN	L100, C, R100	C, L40, R40	Left to right. OSC 1 starts in the middle, the other two to either side.
4 5 6	WIDTH	0 to 100	0	Spreads the oscillator's two detuned halves apart, one left and one right. Wide and shimmering with some DETUNE (page 2); at 0 the oscillator is a single point.
7 8 9	REV SEND	0 to 100	0	How much of the oscillator, echoes included, goes into the reverb. At 0 it stays dry.
10	SIZE	0 to 100	75	The reverb's size and length, from a small room to a long hall.

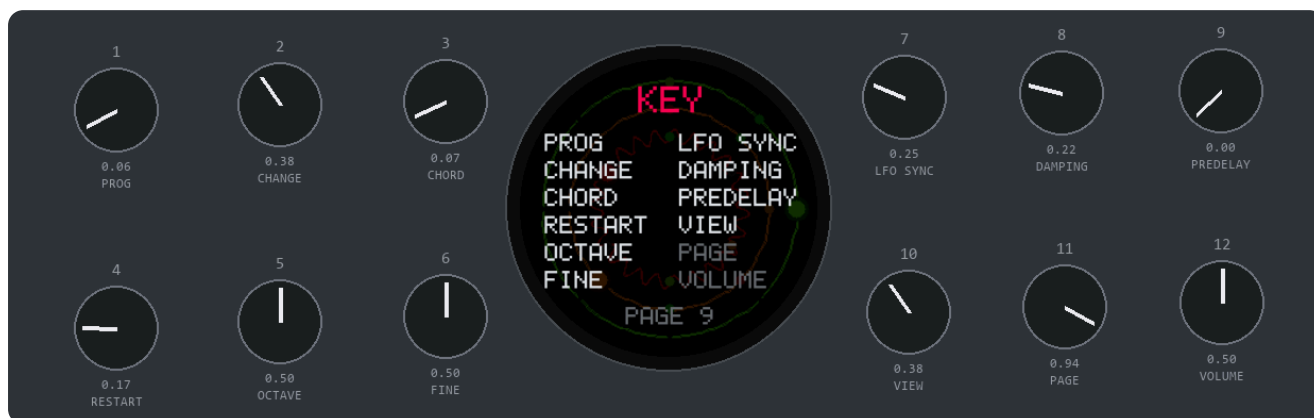
There is one reverb, shared by all three. Its tone is set on page 9 (DAMPING, PREDELAY) and the master REVERB knob on page 1 scales all three sends at once.

TRY THIS

- Send only the high OSC 3 to the reverb, at about 60, with SIZE 90: a halo above a dry, punchy low end.
- DETUNE 15 on page 2 and WIDTH 100 on all three: the power-on patch becomes a wide stereo wash.

KEY

Chord progressions, tuning, a few reverb details, and which picture the screen shows.



Page 9. Like MAIN, a whole-instrument page with a separate job for every knob.

KNOB	FUNCTION	SETTINGS	POWER-ON	WHAT IT DOES
1	PROG	OFF POP 50S SAD EPIC JAZZ VAMP CANON FALL	OFF	A chord progression that steps along by itself (table below). OFF holds the chord set by CHORD.
2	CHANGE	1/2 BAR 1 BAR 2 BARS 4 BARS	1 BAR	How long each chord of the progression lasts. Changes always land on the beat.
3	CHORD	I to VII	I	With PROG OFF: the chord to play, by its place in the key. Play your own progression by turning it.
4	RESTART	FREE BAR CHORD	FREE	When every pattern jumps back to its first step. FREE: never, the patterns run on and drift. BAR: at every bar line. CHORD: at every chord change.
5	OCTAVE	-2 to +2	+0	Moves the whole instrument up or down by octaves.
6	FINE	-50 to +50 CT	+0 CT	Fine tuning, up to a quarter-tone either way, to play along with a record or another instrument.
7	LFO SYNC	FREE TEMPO	FREE	TEMPO locks all fifteen LFOs to the master clock, with RATE picking a note length (page 6).
8	DAMPING	12.0 KHZ to 1.0 KHZ	6.9 KHZ	The reverb's tone: turn right for a darker, softer reverb.
9	PREDELAY	0 to 250 MS	0 MS	A gap before the reverb starts. Keeps notes clear in front of a big reverb.
10	VIEW	RINGS PITCH WHEEL SCOPE	PITCH	What the screen shows (p. 23).

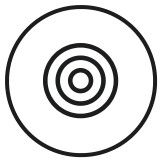
The progressions

PROG	DEGREES	IN A MINOR (POWER-ON)	IN C MAJOR
POP	1 5 6 4	Am Em F Dm	C G Am F
50S	1 6 4 5	Am F Dm Em	C Am F G
SAD	6 4 1 5	F Dm Am Em	Am F C G
EPIC	1 6 3 7	Am F C G	C Am Em Bdim
JAZZ	2 5 1 1	Bdim Em Am Am	Dm G C C
VAMP	1 4	Am Dm	C F
CANON	1 5 6 3 4 1 4 5	Am Em F C Dm Am Dm Em	C G Am Em F C F G
FALL	1 7 6 5	Am G F Em	C Bdim Am G



POP running in A minor, on its third chord: F, degree VI, chord 3 of 4. The four dots are the beats of the bar.

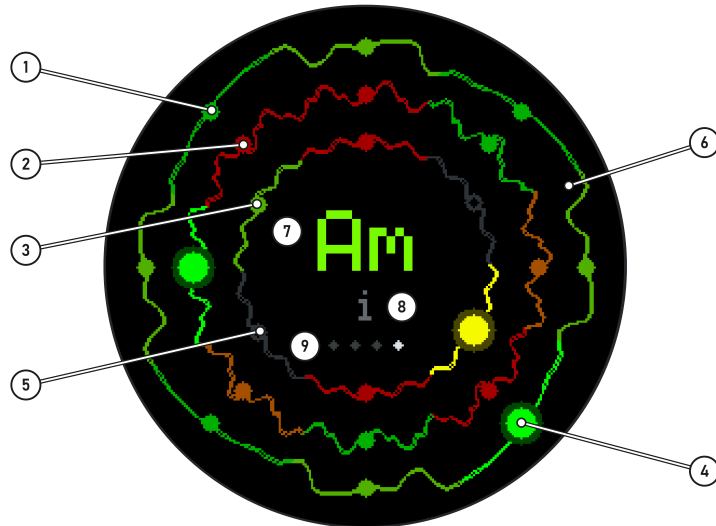
A progression is written in scale degrees, so it works in any key: ROOT and SCALE on page 1 still transpose it while it runs. The chord changes on the bar line; a note already sounding finishes, and the next one uses the new chord. With a chord key (SCALE set to MAJ, MIN7 and so on) every chord of the progression takes that chord's quality.



THE SCREEN

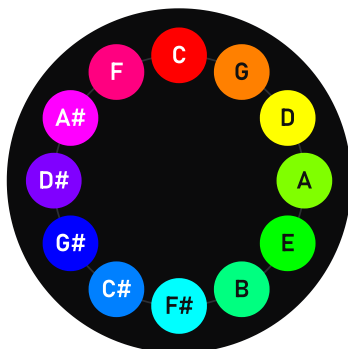
Reading the rings

The screen shows what the three arps are playing, as it happens. Once you can read it, you can see a pattern before you hear it.



- 1 **OSC 1** is the outer ring.
- 2 **OSC 2** is the middle ring.
- 3 **OSC 3** is the inner ring. Each ring is cut into as many arcs as its oscillator's **LENGTH**, starting at 12 o'clock and running clockwise. One arc is one step.
- 4 **The playing step** is the big dot with a halo. It moves on one arc with every step.
- 5 **A rest** is a hollow dot on a grey arc: a step that **DENSITY** leaves out.
- 6 **The line is the sound itself.** It is the oscillator's real waveform: it swells when a note starts and flattens as it fades, and higher notes fit more waves round the ring. A drone is one unbroken ring.
- 7 **The chord** playing now, in the color of its root note.
- 8 **Its numeral** in the key. Lowercase means minor. When a progression runs, its place in it follows, e.g. 3/4.
- 9 **The beat:** one dot for each beat of the bar.

Colors are notes



Every note has its own color, the same in every octave and on every view. The colors run around the *circle of fifths*: notes a fifth apart get neighboring colors.

That puts notes that sound good together close in color. A root and its fifth (the two notes of a **FIFTHS** pool) always look alike, and each chord of a progression brings its own mix of colors to the screen.

The wheel is drawn on black because the brightest colors, yellow and cyan, vanish on white paper. On the screen they glow.

Four views page 9 KEY, knob 10 VIEW

The same music can be drawn four ways. Pick one with **VIEW** ; the readout and page maps look the same on all of them.



PITCH power-on

The rings, with each whole ring in the color of the note its oscillator is playing right now. As the arp moves, the ring changes color: you watch the melody.



RINGS

Each arc keeps the color of the note on its step, so the whole pattern is on screen at once, and you see it change when the chord does.



WHEEL

The twelve notes around the circle of fifths, C at the top. The chord's notes are joined into a shape: its outline shows the kind of chord. Each oscillator's note is a bright dot (biggest for OSC 1), nearer the rim the higher it plays.



SCOPE

The three waveforms drawn over each other, each in its note's color. The plainest view, and the best for hearing what **SHAPE** and the filter do.

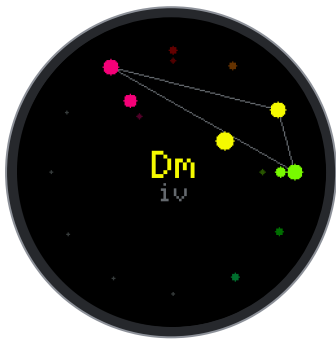


FIVE PATCHES

Recipes

Each recipe starts from the power-on sound, so switch Ammonite off and on first (with knob 11 on page 1). Then make the settings in any order.

The values are exactly what the screen's readout shows when the knob is right. The three columns are OSC 1, 2 and 3; a grey dash means leave that one alone.



Slow tide

Three drones that slide from chord to chord under a slow, breathing filter. No arps at all: the progression does the moving.

PAGE	SECTION	SETTING	OSC 1	OSC 2	OSC 3
1 MAIN		TEMPO	70 BPM		
2 OSC		SHAPE	25	10	0
		DETUNE	12.0 CT		
3 ARP	NOTES	MODE	OFF	OFF	OFF
4 ENVELOPE	PITCH	GLIDE	1.0 S	1.0 S	1.0 S
5 FILTER		CUTOFF	763 HZ	980 HZ	1.3 KHZ
6 LFO	CUTOFF	DEPTH	35	35	35
8 MIX		WIDTH	60	60	60
		REV SEND	60	60	60
		SIZE	90		
9 KEY		PROG	SAD		
		CHANGE	2 BARS		



Clockwork

Three sixteenth-note patterns of 8, 7 and 5 steps, each with a few steps left out. They line up only once every 280 steps, so the groove keeps turning over.

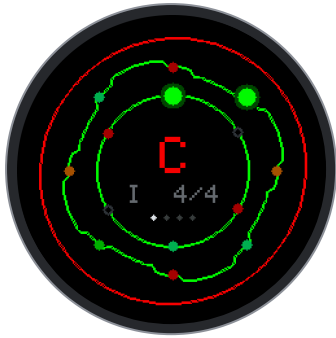
PAGE	SECTION	SETTING	OSC 1	OSC 2	OSC 3
1 MAIN		TEMPO	120 BPM		
3 ARP	NOTES	MODE	UP	RANDOM	UP-DN
		POOL	ROOT	TRIAD	KEY
	RHYTHM	DIVISION	1/16	1/16	1/16
		LENGTH	8	7	5
	CHANCE	DENSITY	5/8	4/7	3/5
4 ENVELOPE	AMP	DECAY	151 MS	151 MS	151 MS
		SUSTAIN	0	–	–
	FILTER	AMOUNT	+2.0 OCT	+2.0 OCT	+2.0 OCT
5 FILTER		CUTOFF	763 HZ	763 HZ	763 HZ
		RESO	50	50	50
7 DELAY	TIME	SEND	–	35	–



Dub echoes

A sparse, resonant chord stab on OSC 1 thrown into a long, darkening, wobbly ping-pong echo, over a sine bass that follows the chords.

PAGE	SECTION	SETTING	OSC 1	OSC 2	OSC 3
1 MAIN		LEVEL	–	0	55
		ROOT	G		
		TEMPO	90 BPM		
2 OSC		SHAPE	50	–	0
		OCTAVE	+1	–	-1
		DEGREE	–	–	+0
3 ARP	NOTES	MODE	RANDOM	–	OFF
	RHYTHM	DIVISION	1/8	–	–
	CHANCE	GATE	20	–	–
		DENSITY	3/8	–	–
4 ENVELOPE	AMP	DECAY	197 MS	–	–
		SUSTAIN	0	–	–
5 FILTER		CUTOFF	1.6 KHZ	–	–
		RESO	55	–	–
		TYPE	BP	–	–
7 DELAY	TIME	FEEDBACK	79	–	–
		SEND	60	–	–
	COLOR	TONE	1.5 KHZ	–	–
		WOBBLE	2.1 MS	–	–
8 MIX		REV SEND	30	–	–
9 KEY		PROG	VAMP		
		CHANGE	2 BARS		



Late set

A swung ii-V-I in C: a gliding bass on the chord roots, an arp through each seventh chord, and a looser triplet line on top.

PAGE	SECTION	SETTING	OSC 1	OSC 2	OSC 3
1 MAIN		ROOT	C		
		SCALE	MAJOR		
		TEMPO	96 BPM		
		SWING	50		
2 OSC		SHAPE	0	30	20
		DEGREE	–	+0	+0
3 ARP	NOTES	MODE	OFF	UP	RANDOM
		POOL	–	7TH	7TH
	RHYTHM	DIVISION	–	1/8	1/4T
	CHANCE	VARY	–	–	30
4 ENVELOPE	AMP	DECAY	–	295 MS	–
		SUSTAIN	–	20	–
8 MIX		REV SEND	–	35	35
9 KEY		PROG	JAZZ		



Bells in D

Pachelbel's chords in D major, played by sine bells: every note starts an octave high and drops into place in a few milliseconds, which is what makes it ring.

PAGE	SECTION	SETTING	OSC 1	OSC 2	OSC 3
1 MAIN		ROOT	D		
		SCALE	MAJOR		
		TEMPO	80 BPM		
2 OSC		SHAPE	30	0	0
3 ARP	NOTES	MODE	OFF	UP	DOWN
	CHANCE	DENSITY	–	–	5/6
4 ENVELOPE	AMP	DECAY	–	917 MS	917 MS
	PITCH	AMOUNT	–	+12 ST	+12 ST
		DECAY	–	8 MS	8 MS
7 DELAY	TIME	SEND	–	–	30
8 MIX		REV SEND	–	45	45
		SIZE	85		
9 KEY		PROG	CANON		
		CHANGE	1/2 BAR		
		RESTART	CHORD		



Scales and chords

Every SCALE setting on page 1, spelled out from C. Turn ROOT and the same pattern starts from another note.

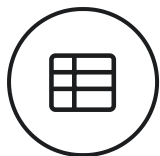
SCALES

SCALE	NOTES FROM C	CHARACTER
MAJOR	C D E F G A B	bright, settled
MINOR	C D D# F G G# A#	dark; the power-on key
DORIAN	C D D# F G A A#	minor with a lift; soul and jazz
PHRYGIAN	C C# D# F G G# A#	dark and tense, Spanish flavor
LYDIAN	C D E F# G A B	major, floating and dreamy
MIXO	C D E F G A A#	major with a bluesy seventh
HARM MIN	C D D# F G G# B	minor with a dramatic leading note
PENTA	C D E G A	five notes, no clashes at all
PENTA MI	C D# F G A#	the minor five-note scale
BLUES	C D# F F# G A#	minor pentatonic plus the blue note

CHORD KEYS

Past BLUES, SCALE turns into a chord. Then POOL KEY plays that chord's notes, and a progression moves the chord through its parent scale (natural minor for MIN, MIN7 and DIM; major for the rest), keeping its quality: a MIN7 key gives a progression of minor sevenths.

SCALE	NOTES FROM C	CHARACTER
MAJ	C E G	plain major triad
MIN	C D# G	plain minor triad
SUS2	C D G	open, neither major nor minor
SUS4	C F G	open, waiting to resolve
MAJ7	C E G B	soft, jazzy major
MIN7	C D# G A#	smooth minor
DOM7	C E G A#	bluesy, restless
DIM	C D# F#	tense, spooky
AUG	C E G#	strange, dreamlike
ADD9	C D E G	bright and airy



APPENDIX

Cheat sheet

Every knob on every page. Knob 11 is always PAGE, knob 12 always VOLUME.

PAGE	SECTION	KNOBBS 1-3	KNOBBS 4-6	KNOBBS 7-9	KNOB 10
1 MAIN	whole instrument	1-3 LEVEL	4 ROOT · 5 SCALE · 6 TEMPO	7 CUTOFF · 8 DELAY · 9 REVERB	SWING
2 OSC		SHAPE	OCTAVE	DEGREE	DETUNE
3 ARP	NOTES	MODE	POOL	RANGE	SECTION
	RHYTHM	DIVISION	LENGTH	GATE	
	CHANCE	DENSITY	VARY	RATCHET	
4 ENVELOPE	AMP	ATTACK	DECAY	SUSTAIN	SECTION
	FILTER	ATTACK	DECAY	AMOUNT	
	PITCH	AMOUNT	DECAY	GLIDE	
5 FILTER		CUTOFF	RESO	TYPE	DRIVE
6 LFO	PITCH · CUTOFF · AMP · PAN · SHAPE	RATE	DEPTH	WAVE	TARGET
7 DELAY	TIME	DIVISION	FEEDBACK	SEND	SECTION
	COLOR	TONE	WOBBLE	WIDTH	
8 MIX		PAN	WIDTH	REV SEND	SIZE
9 KEY	whole instrument	1 PROG · 2 CHANGE · 3 CHORD	4 RESTART · 5 OCTAVE · 6 FINE	7 LFO SYNC · 8 DAMPING · 9 PREDELAY	VIEW

On pages 2 to 8 each row's three knobs are OSC 1, 2 and 3, left to right, in both groups. Turning knob 10 on a page with sections, or knob 11, never changes the sound; a knob takes over (and its setting jumps to it) when you turn it.